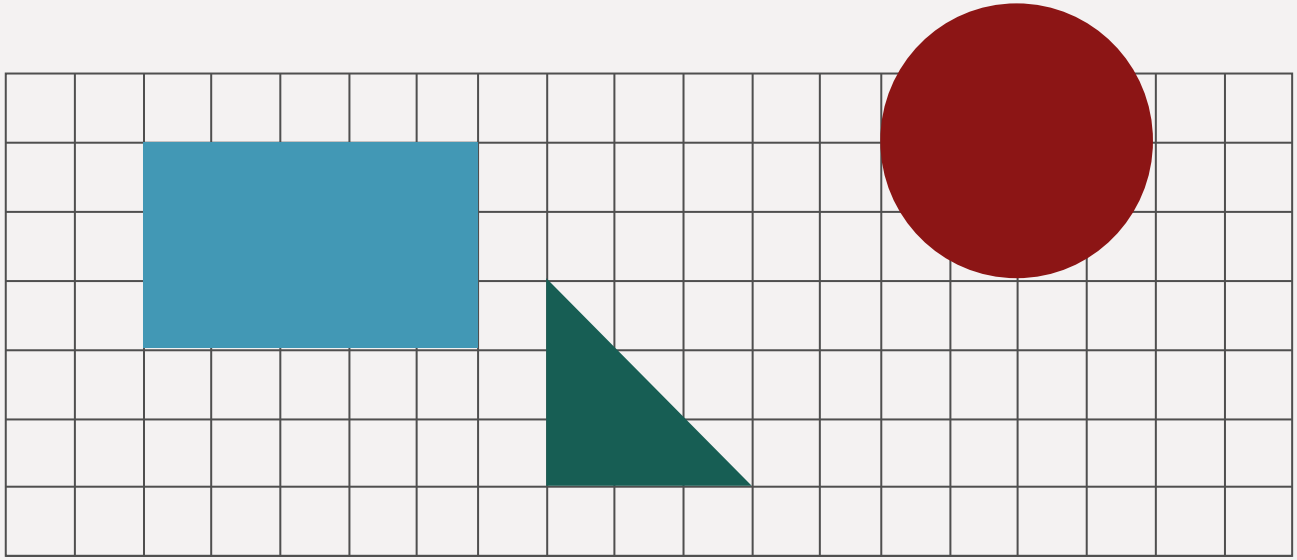
Same voice, different labels:
Effects of ambient talker voices on social categorization

Grace Brown & Meghan Sumner | Stanford University
NWAV 54
October 23, 2026



(Eckert 2008; Irvine 2001; Pharao et al. 2014)

Social meaning is relational

We should see evidence for this in perception:

"Modern Copenhagen"

feminine

gay

immigrant

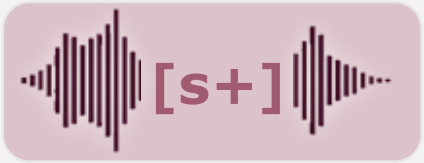


"Street Style"

feminine

gay **gangster**

immigrant



A linguistic variant can mean different things depending on the acoustic features around it

(Eckert 2008; Irvine 2001; Campbell-Kibler 2009; Levon 2007; Pharao et al. 2014; Jessee & Calder 2025; Brown et al. 2026)

Social meaning is relational

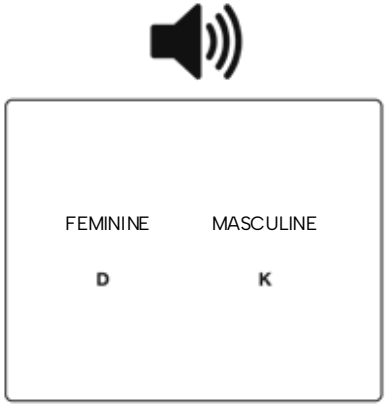
Evidence in perception:

- linguistic cues
- acoustic cues
- listener experiences
- listener ideologies

Does a voice's perceived social meaning depend on the voices around it?

How do the voices that accompany a talker shape the way their voice is socially categorized?

Speeded forced-choice task



feminine
masculine

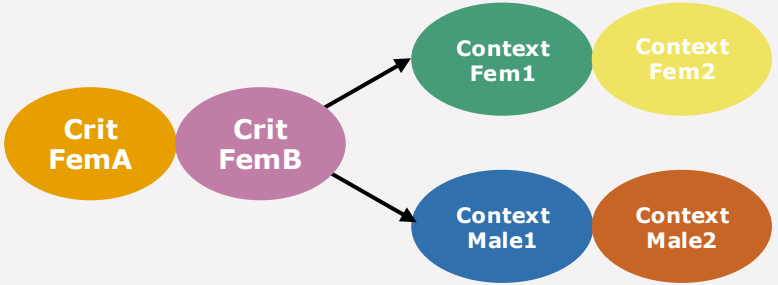
comfortable
awkward

sharp
dull

emotive
robotic

Measures
Response Label (group & participant means)
Response Time (log-rt)

Between Subj. Manipulation



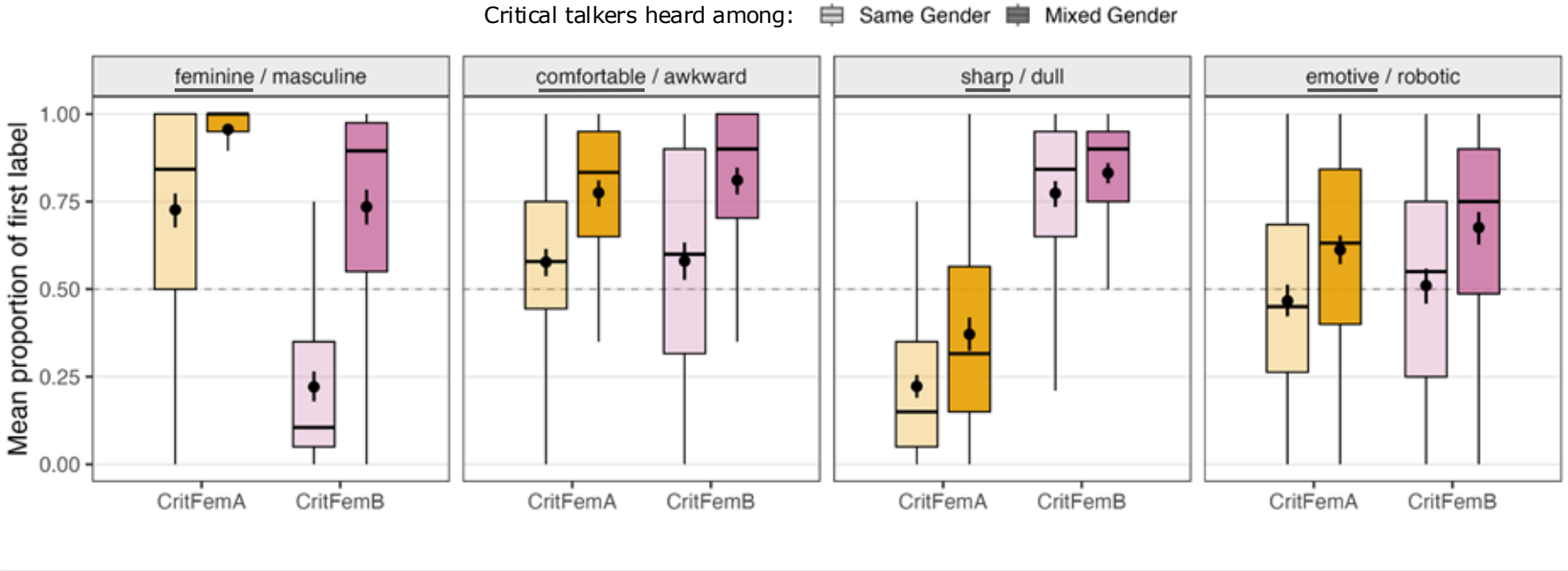
N = 284 participants; 40 trials x 8 blocks = 320 trials

Critical talkers' speech held constant

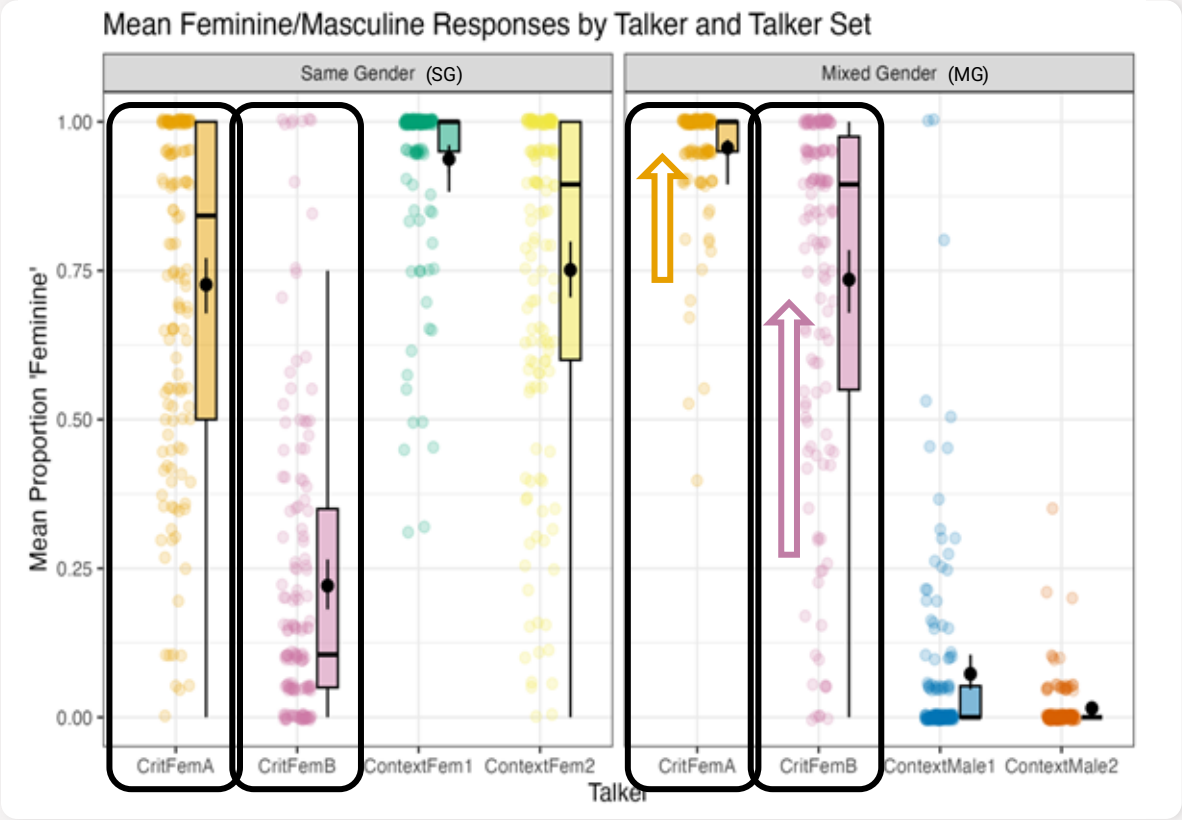
→ **Any difference in categorization behavior must come from accompanying talkers, not the signal**

How does social category assignment for "critical talkers" vary by company?

Talker set is a significant predictor of response proportions for **all label pairs**



Company changes which label a voice receives & how consistently that label is assigned



CritFemA

near-categorical "feminine" in MG

CritFemB

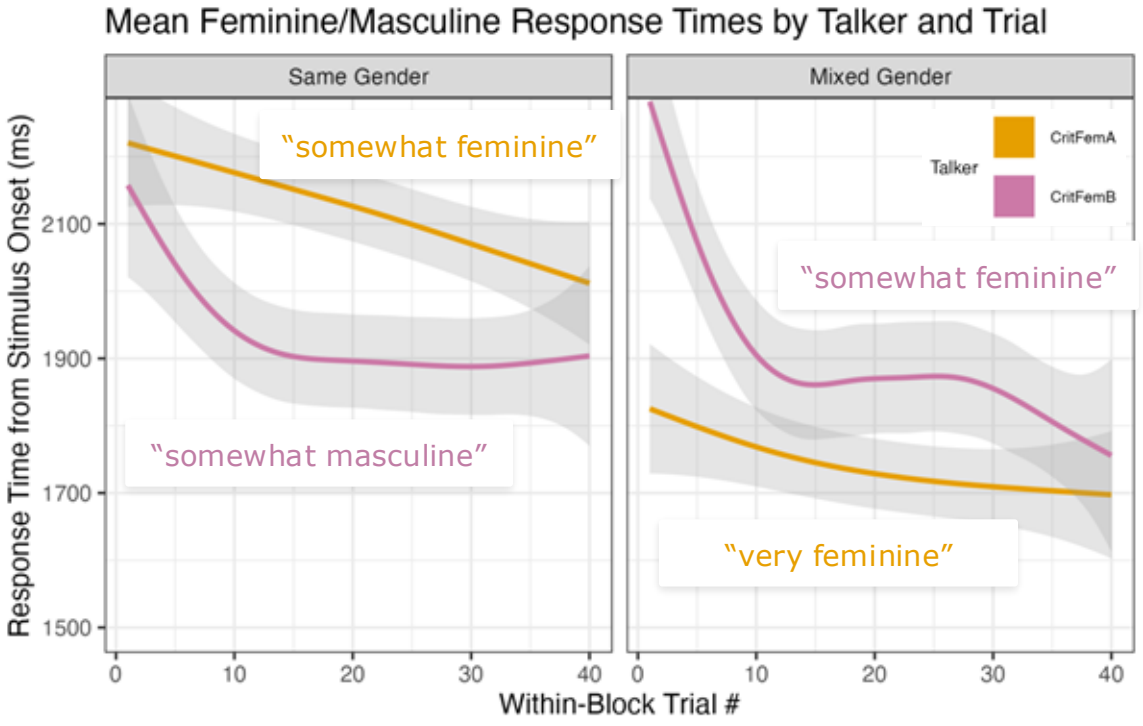
"masculine" vs "feminine"
0.23 **0.74**

"feminine" **>20x more likely** in MG

Relative ordering of
CritFemA > CritFemB
preserved across both conditions

What is the time course of these decisions?

Company can change how quickly a voice is categorized



CritFemA

Label stable ▪ RT differs

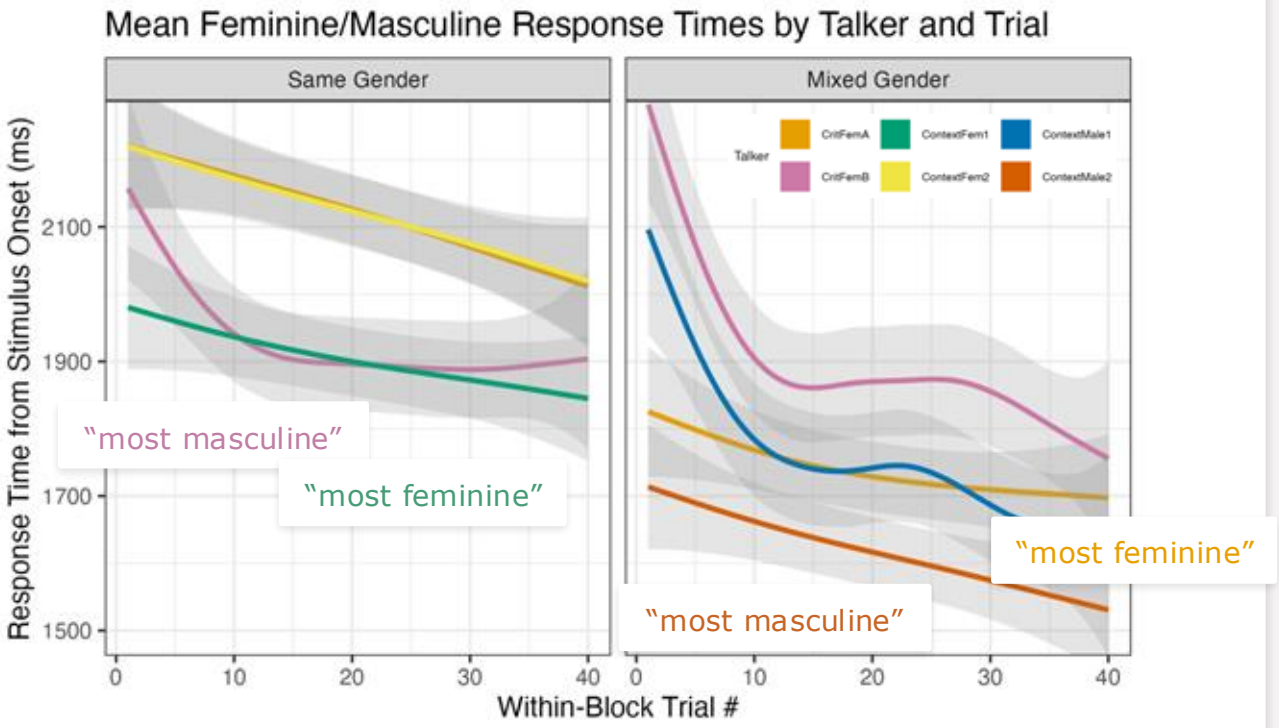
Stronger activation in MG → faster RT

CritFemB

Label differs ▪ RT stable

Similar activation of "masculine" in SG & "feminine" in MG → similar RT

Company can change how quickly a voice is categorized… this corresponds with goodness-of-fit



Relative RT ordering reversal

Same Gender:
CritFemA RT > CritFemB RT

Mixed Gender:
CritFemA RT < CritFemB RT

Within and across participants:
RT correlates with categorization decision consistency

Faster RT → Easier categorization decision

Company Effect
Changing the voices that accompany a talker impacts how they are perceived

Categorization Decision

Same person can receive different social labels depending on who they're heard with

Category Activation

Consistency of social labeling and response time depend on what other voices are present

(e.g., Koffka 1935; Parducci 1965; Sedivy et al. 1999; Kim & Sumner 2015; Casasanto & Lupyan 2015)

We do this relative processing all the time!

lexical access

basic sensory perception (size, loudness, color, etc.)

referential language use

…and more!

Company effects reflect a domain-general process

Boersma, P., & Weenink, D. (2025). "Praat: Doing phonetics by computer," https://praat.org/.
Brown, G., Sumner, M., & Podesva, R. (2026). Indexical weight is distributed: Evidence from social evaluations of /s/. Proceedings of the Linguistic Society of America, 11(1), 6045.
Campbell-Kibler, K. (2009). The nature of sociolinguistic perception. Language Variation and Change, 21(1), 135–156.
Casasanto, D., & Lupyan, G. (2015). "All Concepts Are Ad Hoc Concepts," in E. Margolis & S. Laurence (Eds.), *The Conceptual Mind: New Directions in the Study of Concepts* (pp. 543–566). The MIT Press.
de Leeuw, J., Gilbert, R., & Luchterhandt, B. (2023). jsPsych: Enabling an Open-Source Collaborative Ecosystem of Behavioral Experiments, *Journal of Open Source Software 8*, 5351.
Eckert, P. (2008). Variation and the indexical field, *Journal of Sociolinguistics 12*, 453–476.
Jessee, E., & Calder, J. (2025). The Cisgender Listening Subject in Sociolinguistic Perception: Transgender Identity Affects Sibilant Categorization in American English. Journal of Sociolinguistics, 29(3), 168–181.
Kim, S. K., & Sumner, M. (2015). "Effects of emotional prosody and attention on semantic priming," in *Proceedings of the Annual Meeting of the Cognitive Science Society,* Vol. 37.
Koffka, K. (1935). *Principles of Gestalt psychology*. Harcourt, Brace.
Levon, E. (2014). Categories, stereotypes, and the linguistic perception of sexuality. Language in Society, 43(5), 539–566.
Ochs, E. (1991). "Indexing gender," in A. Duranti & C. Goodwin (Eds.), *Rethinking Context: Language as an Interactive Phenomenon* (pp. 335–358). Cambridge University Press, Cambridge.
Parducci, A. (1965). Category judgment: A range-frequency model, *Psychological Review 72*(6), 407–418.
Pharao, N., Maegaard, M., Møller, J. S., & Kristiansen, T. (2014). Indexical meanings of [s+] among Copenhagen youth: Social perception of a phonetic variant in different prosodic contexts. Language in Society, 43(1), 1–31.
Sedivy, J. C., K. Tanenhaus, M., Chambers, C. G., and Carlson, G. N. (1999). Achieving incremental semantic interpretation through contextual representation, *Cognition 71(*2), 109–147.

THANK YOU

